

The empirical recurrence for OEIS A253332: recovering a conjecture that is not written down

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Abstract

OEIS A253332 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 3249$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A253332 is “Number of $(7+1)X(n+1)$ 0..1 arrays with every $2X2$ subblock sum nondecreasing horizontally, vertically and antidiagonally ne-to-sw.”. It has offset 1 and begins

3249, 9228, 25659, 74953, 210417, 600020, 1660088, 4590122, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:58:31, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 3249$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 3249$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 6498$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	4	c_{23}	−362661863382	c_{45}	505005124022100	c_{67}	3027697305152
c_2	52	c_{24}	167660892763	c_{46}	−833427927434700	c_{68}	10757165906240
c_3	−226	c_{25}	1206296434070	c_{47}	−519968917201500	c_{69}	−1584645895040
c_4	−1296	c_{26}	−699867921970	c_{48}	946891743329025	c_{70}	−3561984436608
c_5	6210	c_{27}	−3567332267996	c_{49}	479133467560800	c_{71}	656835409152
c_6	20530	c_{28}	2453509588140	c_{50}	−974196663714144	c_{72}	1015455091456
c_7	−110578	c_{29}	9422358831360	c_{51}	−392664169605474	c_{73}	−222028538368
c_8	−230665	c_{30}	−7427473197472	c_{52}	907080060441888	c_{74}	−245561625088
c_9	1434490	c_{31}	−22308920543192	c_{53}	283393933688106	c_{75}	61479009280
c_{10}	1934946	c_{32}	19729568725309	c_{54}	−763493755491974	c_{76}	49386654720
c_{11}	−14450904	c_{33}	47481274209668	c_{55}	−177154767356410	c_{77}	−13818415104
c_{12}	−12320932	c_{34}	−46458179718732	c_{56}	579953785268345	c_{78}	−8037591040
c_{13}	117668356	c_{35}	−91032789624030	c_{57}	92946206692618	c_{79}	2469130240
c_{14}	58057636	c_{36}	97659689484240	c_{58}	−396657087078510	c_{80}	1016983552
c_{15}	−795930460	c_{37}	157441549620654	c_{59}	−38010255685740	c_{81}	−338403328
c_{16}	−174529485	c_{38}	−184176843019650	c_{60}	243547963987844	c_{82}	−93855744
c_{17}	4561654968	c_{39}	−245819754993150	c_{61}	9173596634344	c_{83}	33472512
c_{18}	5783960	c_{40}	312733177780725	c_{62}	−133745006323448	c_{84}	5619712
c_{19}	−22482351710	c_{41}	346519749882150	c_{63}	2016114146384	c_{85}	−2129920
c_{20}	4004317408	c_{42}	−479333910398850	c_{64}	65383883242704	c_{86}	−163840
c_{21}	96382920998	c_{43}	−440729463376800	c_{65}	−4126091055840	c_{87}	65536
c_{22}	−31815344106	c_{44}	664327503414900	c_{66}	−28291252127840		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A253332.
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